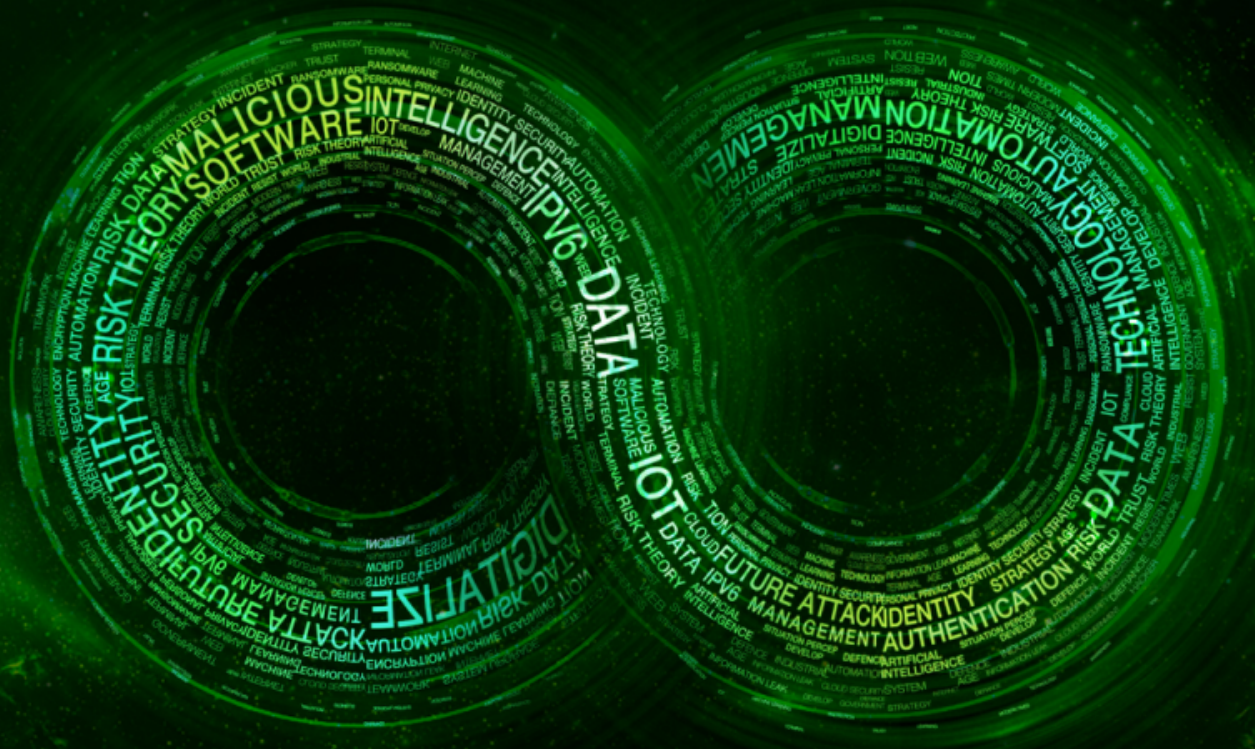




ISC 互联网安全大会



360 互联网安全中心



INTERNET THREATS TO INDUSTRIAL ORGANIZATIONS

互联网威胁正在盯上工业企业

Evgeny Goncharov Head of Kaspersky Lab ICS CERT

ISC 互联网安全大会 中国·北京

Internet Security Conference 2018 Beijing · China



OUR INDUSTRIAL CYBERSECURITY ACTIVITIES IN A NUTSHELL



KICS is a portfolio of technologies and services designed to secure Industrial Control System environments

Launched at the end of 2015

More than 15 customers globally since launch

4 official references



KL ICS Cyber Emergency Response Team is a special non-commercial project that offers a wide range of information and research services
Officially launched at the end of 2016

Regularly contributes to Industrial cybersecurity, e.g. helping prepare IIFS v 1.0

More than **160 ICS** vulnerabilities found

Raising awareness through hackathons, CTF, Workshops and demos worldwide



ISC 互联网安全大会



360 互联网安全中心

Contents 目录

KEY INDUSTRIAL CYBER RISKS

主要的工业网络风险

ICS CYBER RISK MAIN FACTORS

ICS网络风险主要因素

ZERO TRUST SECURITY

KASPERSKY

WEB INTERNET
INFORMATION LEAK
TERMINAL AGE
PERSONAL PRIVACY
IDENTITY SECURITY
IDENTITY
AUTHENTICATION
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China
INDUSTRIAL

KEY ICS SECURITY RISKS: ICS VULNERABILITIES

vulnerabilities in ICS products posing cyber-physical threat



ISC 互联网安全大会



360 互联网安全中心



ZERO TRUST SECURITY



WEB INTERNET
INFORMATION LEAK
MINAL AGE
NATIONAL PRIVACY
IDENTITY SECURITY
AUTHENTICATION
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China
INDUSTRIAL

KEY ICS SECURITY RISKS: ICS VULNERABILITIES

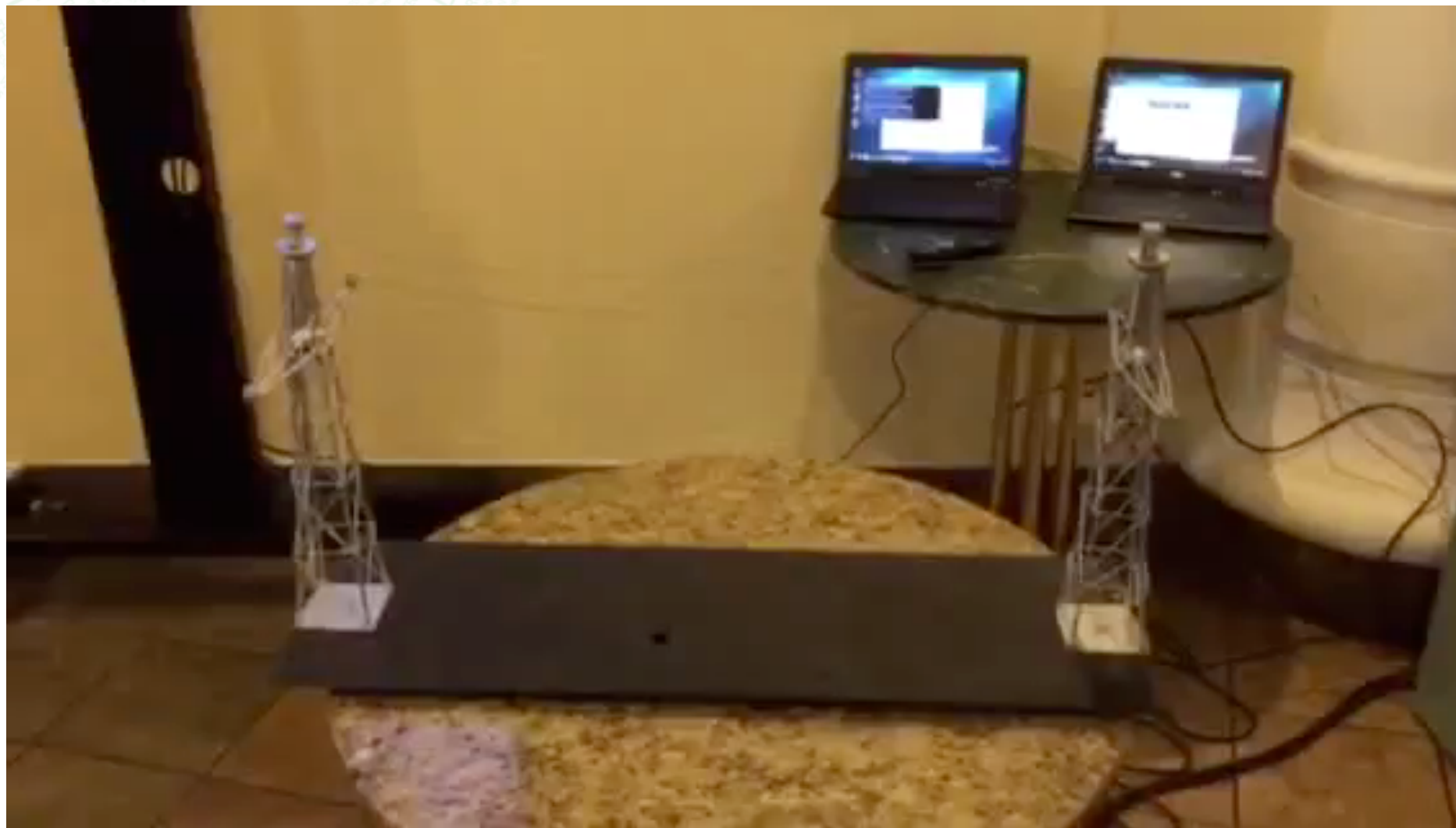
vulnerabilities in ICS products posing cyber-physical threat



ISC 互联网安全大会



360 互联网安全中心



ZERO TRUST SECURITY

KASPERSKY

ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing · China

KEY ICS SECURITY RISKS: ICS VULNERABILITIES

vulnerabilities in ICS products posing cyber-physical threat



ISC 互联网安全大会



360 互联网安全中心



ZERO TRUST SECURITY

KASPERSKY

ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing · China

KEY ICS SECURITY RISKS: ICS VULNERABILITIES

vulnerabilities in ICS products posing cyber-physical threat



ISC 互联网安全大会



360 互联网安全中心



ZERO TRUST SECURITY

KASPERSKY

ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China

KEY ICS SECURITY RISKS: ICS VULNERABILITIES

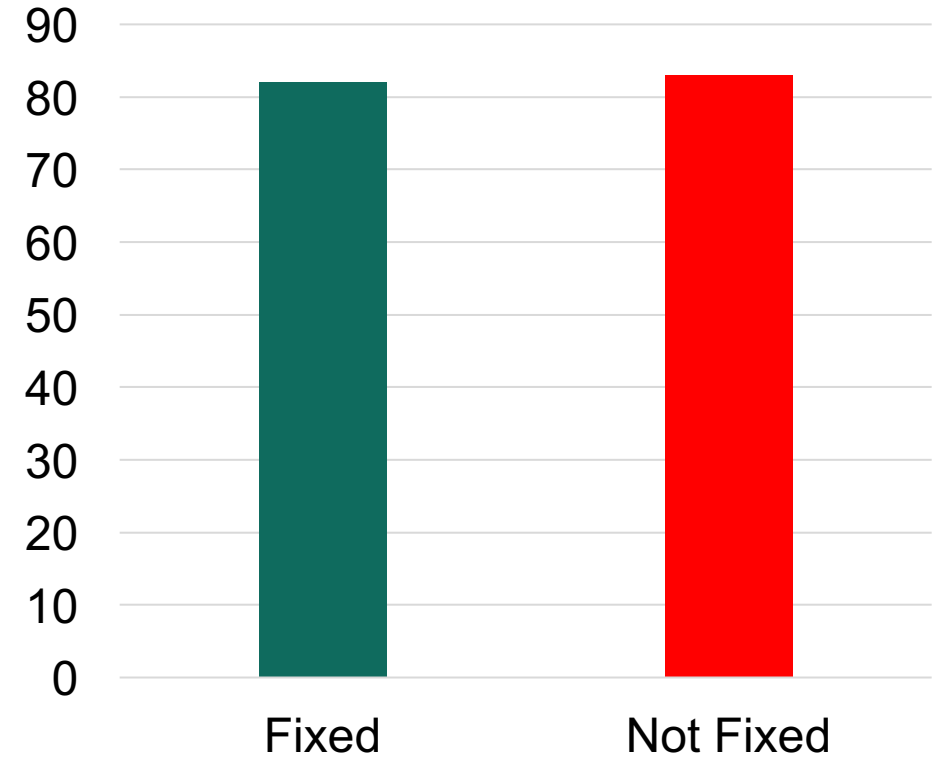
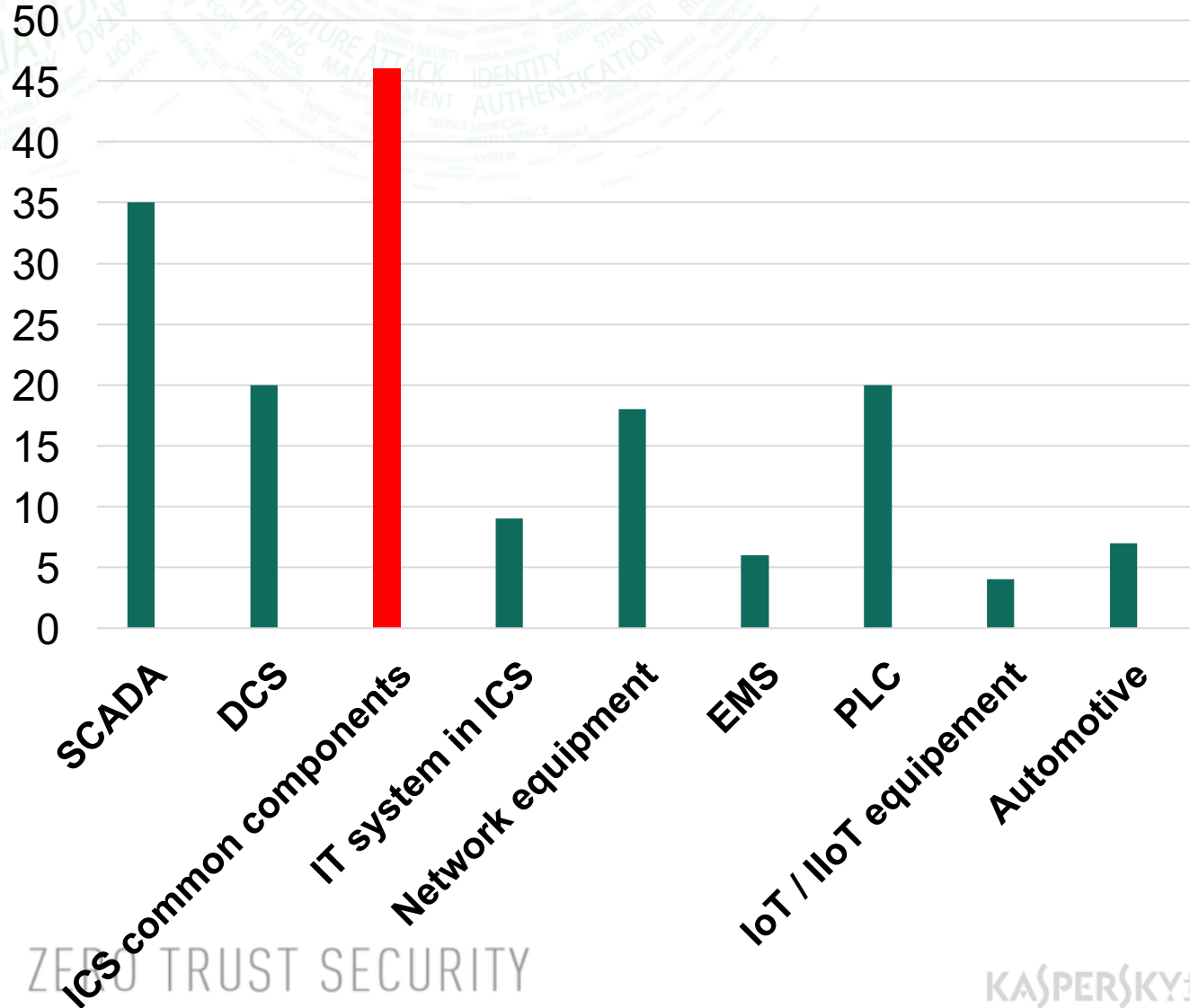
0-day vulnerabilities in ICS products found and reported by KL ICS CERT



ISC 互联网安全大会



360 互联网安全中心



ICS TRUST SECURITY

KASPERSKY

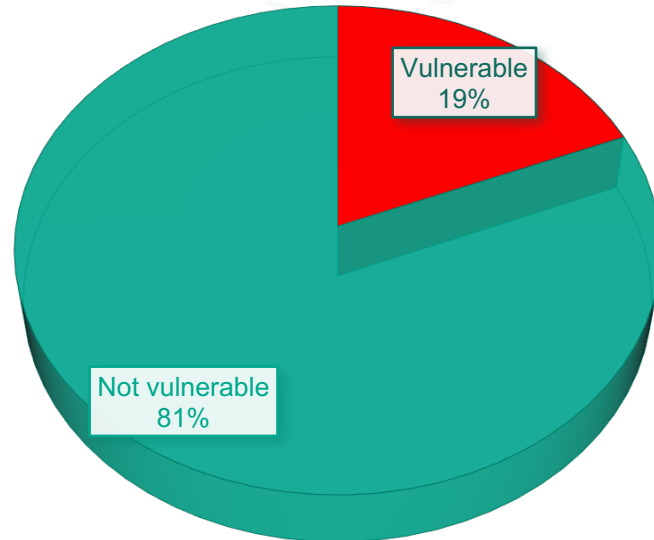
WEB INTERNET
INFORMATION LEAK
TERMINAL AGE
PERSONAL PRIVACY
IDENTITY SECURITY
IDENTITY
AUTHENTICATION
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing · China
INDUSTRIAL

KEY ICS SECURITY RISKS: OUTDATED ICS PRODUCTS

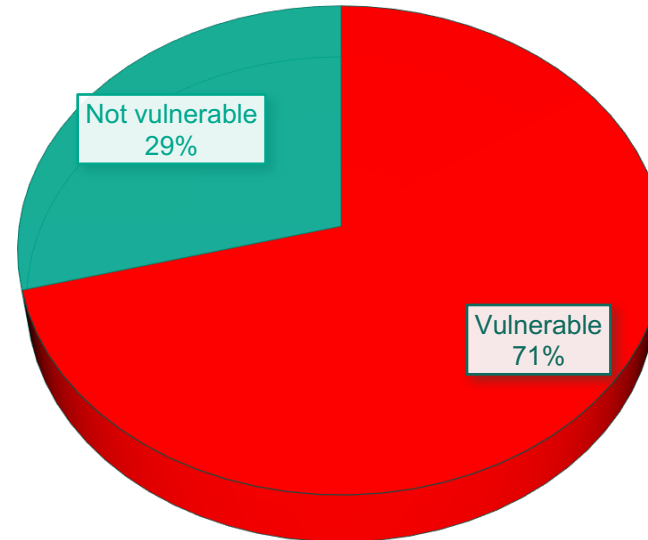


%of vulnerable ICS product installations according to KSN statistics

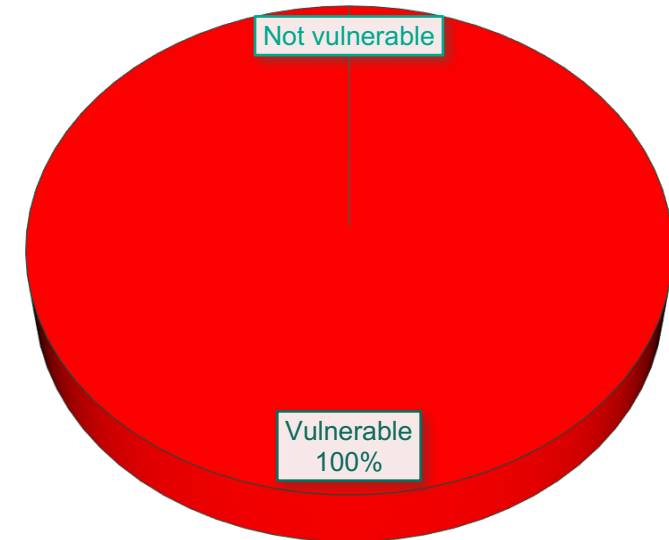
CVE-2013-2811



CVE-2014-0750



CVE-2016-2277



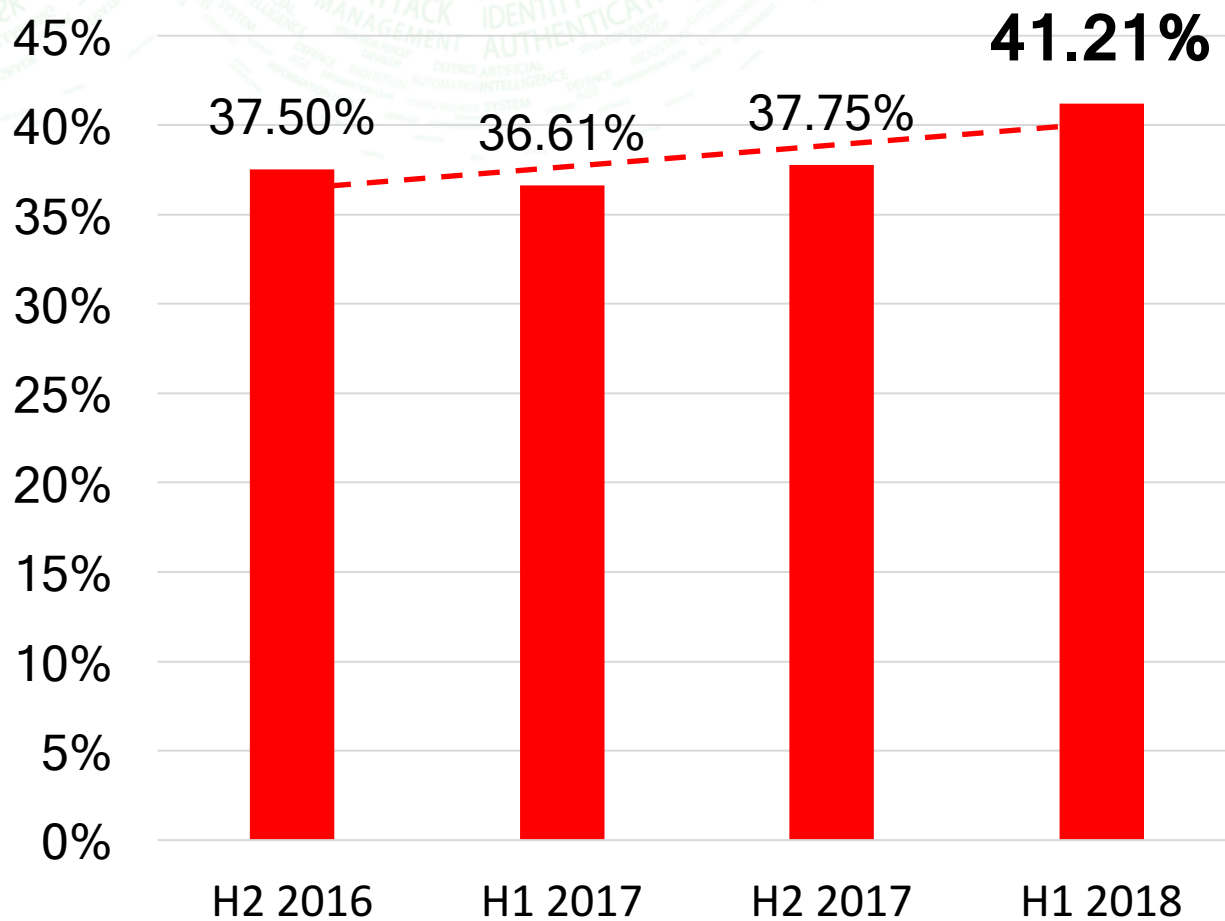
ZERO TRUST SECURITY

KASPERSKY

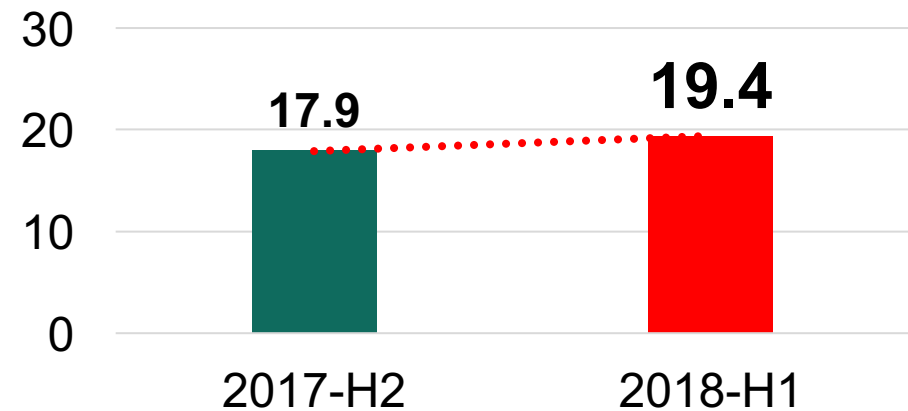
WEB INTERNET
INFORMATION LEAK
TERMINAL AGE
PERSONAL PRIVACY
IDENTITY SECURITY
IDENTITY
AUTHENTICATION
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China
INDUSTRIAL

KEY ICS SECURITY RISKS: RANDOM MALWARE ATTACKS

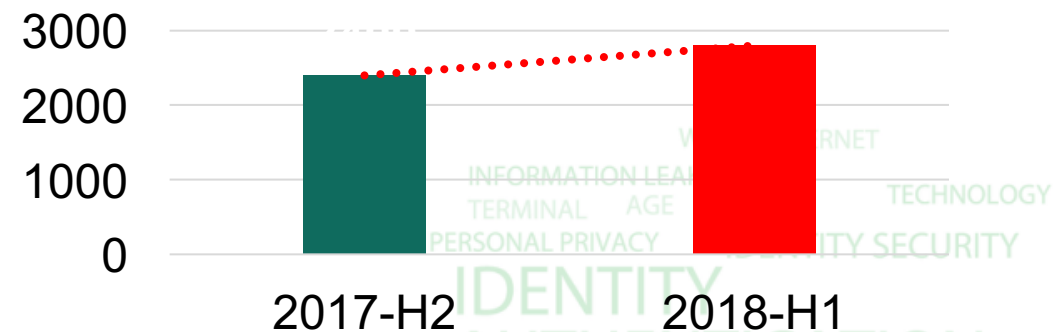
%of ICS computers attacked by malware – according to KSN



Malware modifications



Malware families



KEY ICS SECURITY RISKS: MASS-TARGETTING CAMPAIGNS

Wide-spread attacks targeting hundreds of Industrial organizations



Criminal attack campaigns 犯罪攻击活动

Nigerian phishing: 500+ Industrial organizations world-wide

RU campaign: 400+ Industrial organizations in Russia

Politically motivated (?) attack campaigns 政治动机（？）攻击活动

Energetic Bear / Crouching Yeti attacks in 2017-2018

It's getting harder and harder to find differences

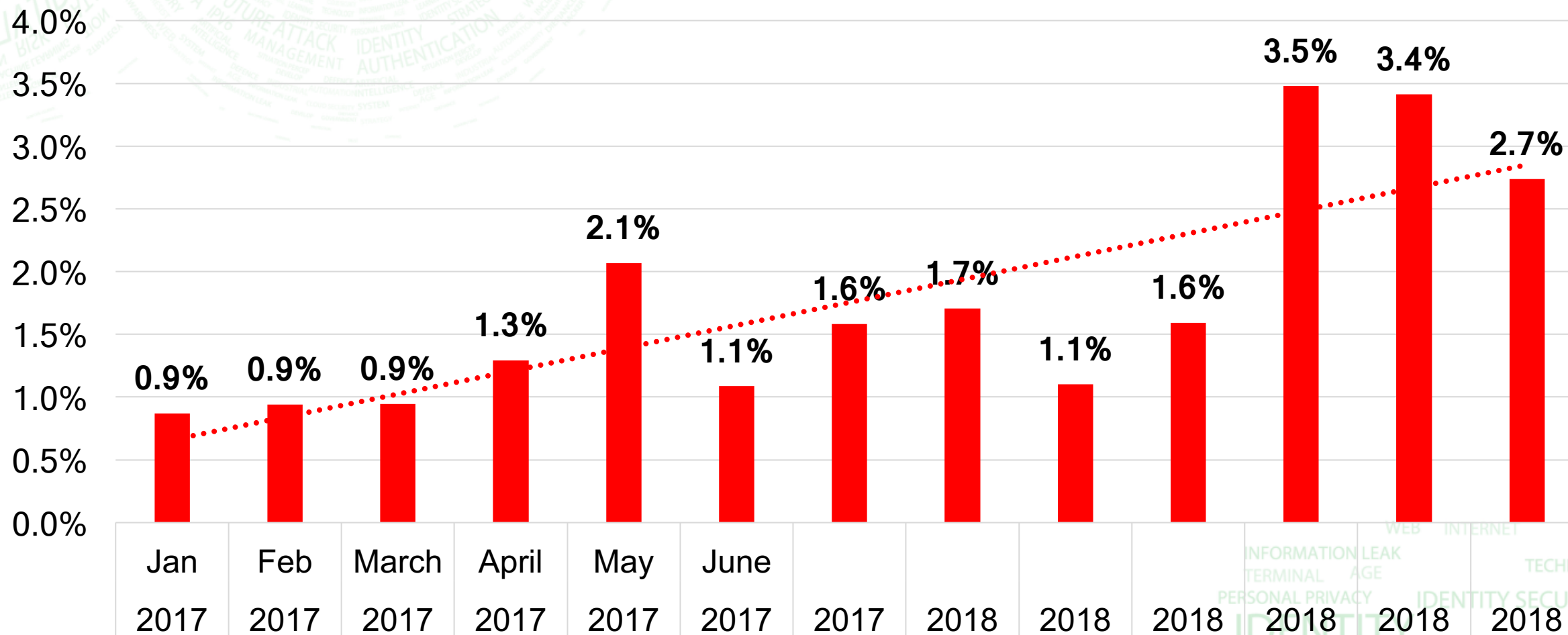
ZERO TRUST SECURITY

KASPERSKY

WEB INTERNET
INFORMATION LEAK
TERMINAL AGE
PERSONAL PRIVACY
IDENTITY SECURITY
IDENTITY
AUTHENTICATION
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China
INDUSTRIAL

KEY ICS SECURITY RISKS: CRYPTO-MINERS HYPE

% of ICS computers attacked by miners - according to KSN statistics



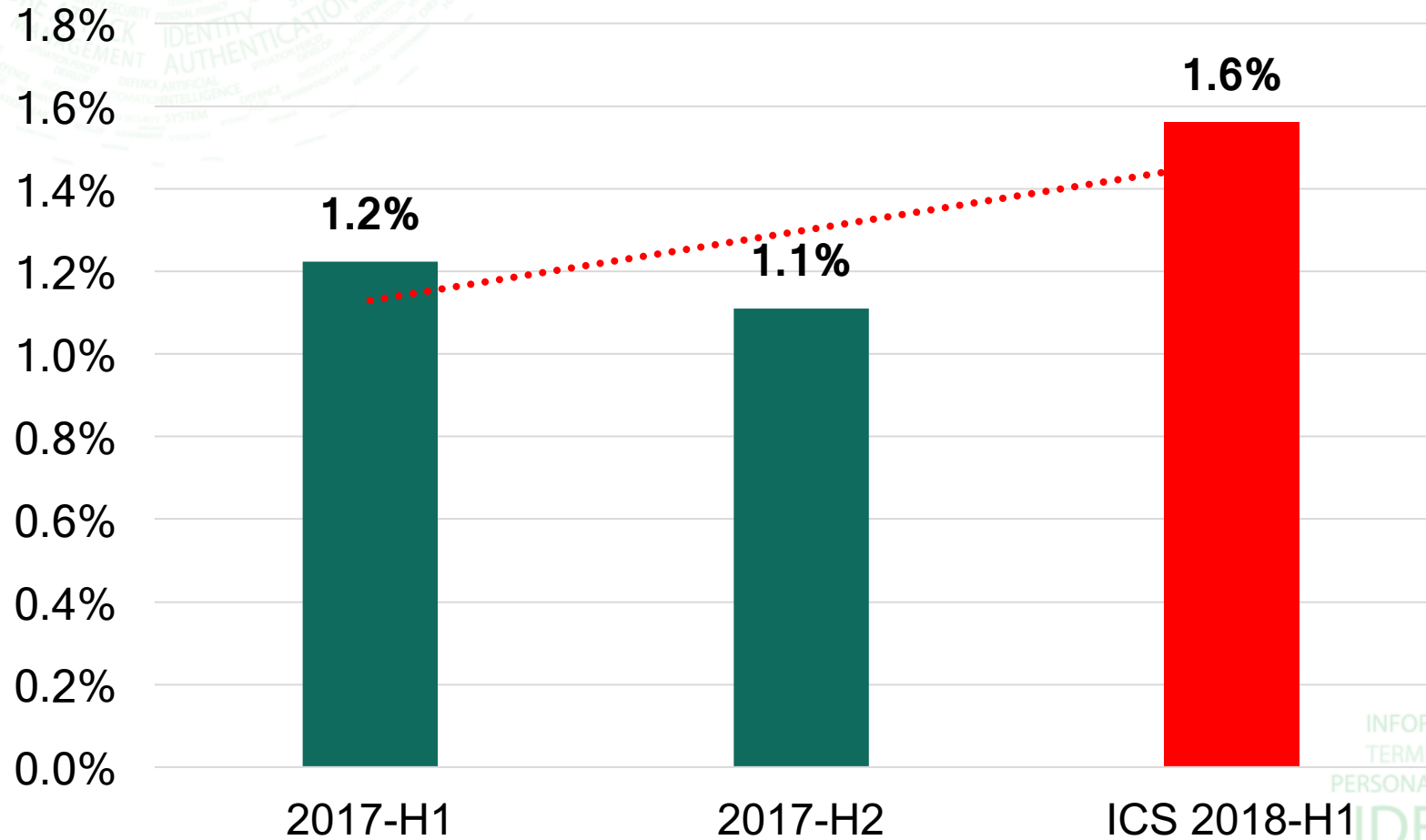
ZERO TRUST SECURITY

KASPERSKY

ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing · China

KEY ICS SECURITY RISKS: RANSOMWARE IN ICS

% of ICS computers attacked by ransomware - according to KSN statistics



ZERO TRUST SECURITY

KASPERSKY

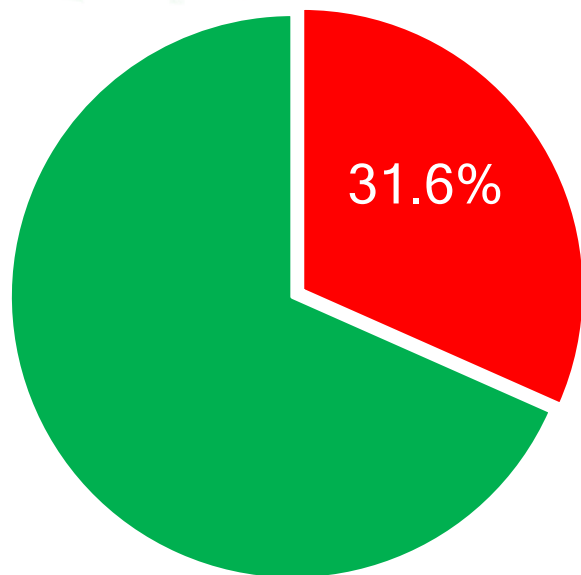
WEB INTERNET
INFORMATION LEAK
TERMINAL AGE
PERSONAL PRIVACY
IDENTITY SECURITY
IDENTITY
AUTHENTICATION
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China
INDUSTRIAL

KEY ICS SECURITY RISKS: LEGITIMATE BACKDOORS

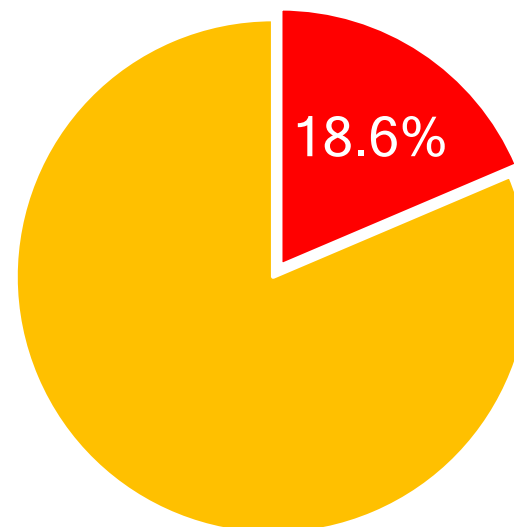
% of ICS computers running remote administration tools (RATs) - according to KSN statistics



% ICS running RAT*



% of ICS computers running RAT* installed with ICS software



*not counting RATs built in Windows OS

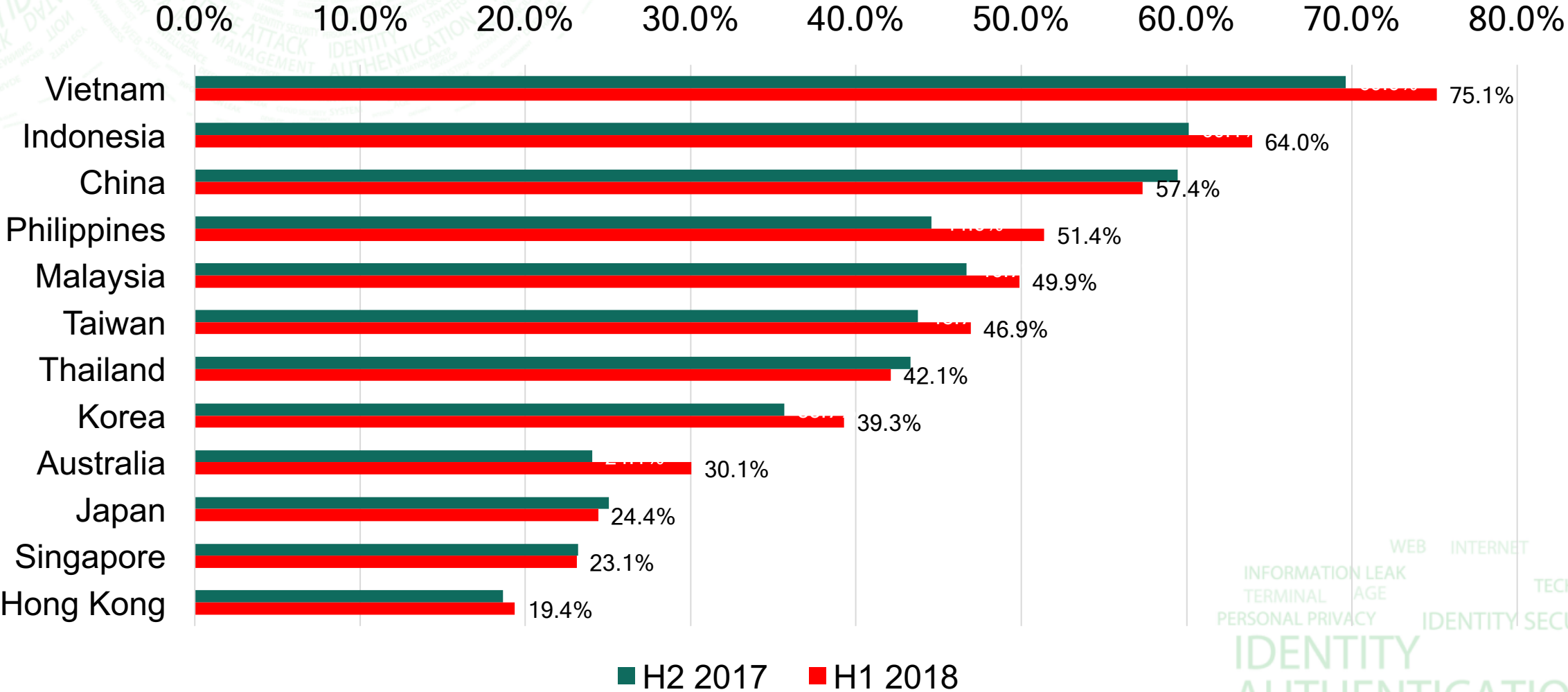
ZERO TRUST SECURITY

KASPERSKY

WEB INTERNET
INFORMATION LEAK
TERMINAL AGE
PERSONAL PRIVACY
IDENTITY SECURITY
IDENTITY
AUTHENTICATION
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China
INDUSTRIAL

RANDOM ATTACKS ON ICS: CHINA VS REGION

%of ICS computers attacked by malware – according to KSN statistics



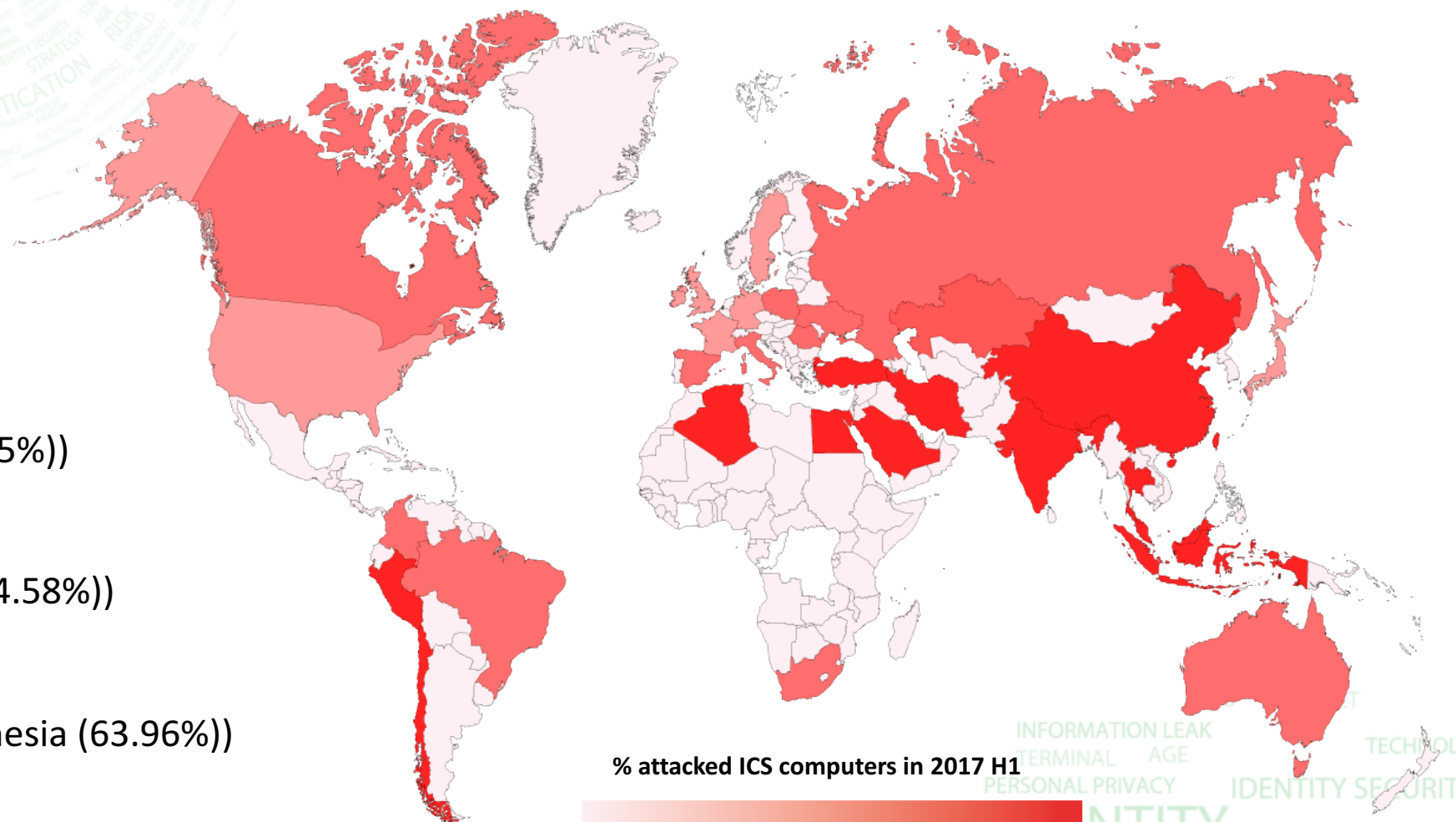
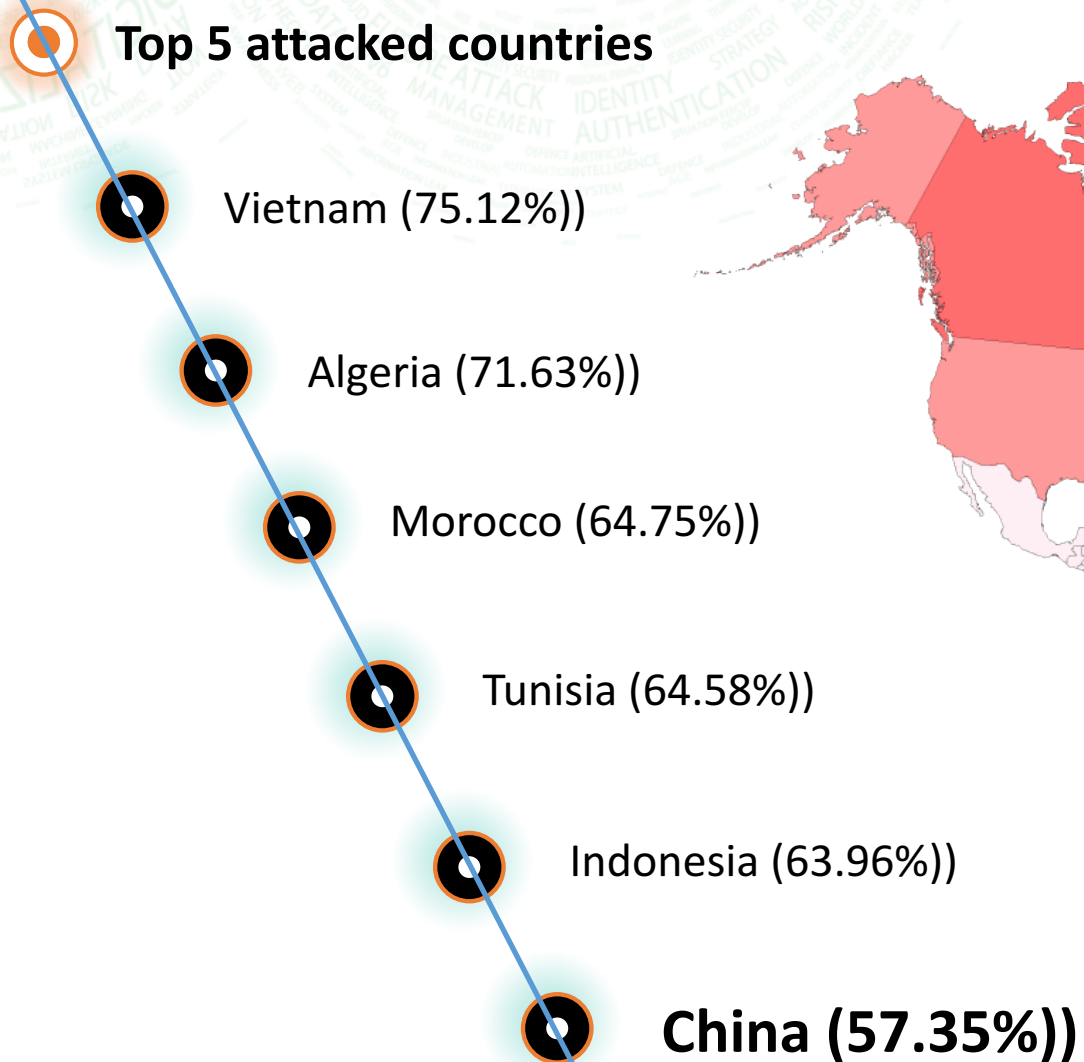
ZERO TRUST SECURITY



WEB INTERNET
INFORMATION LEAK
TERMINAL AGE
PERSONAL PRIVACY
IDENTITY SECURITY
IDENTITY AUTHENTICATION
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China
INDUSTRIAL

RANDOM ATTACKS ON ICS: CHINA VS WORLD

%of ICS computers attacked by malware – according to KSN statistics



ZERO TRUST SECURITY

KASPERSKY

INFORMATION LEAK
TERMINAL AGE
PERSONAL PRIVACY
IDENTITY SECURITY
TECHNOLOGY
AUTHENTICATION
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China
INDUSTRIAL



ISC 互联网安全大会



360 互联网安全中心

Contents 目录

KEY INDUSTRIAL CYBER RISKS

主要的工业网络风险

ICS CYBER RISK MAIN FACTORS

ICS网络风险主要因素

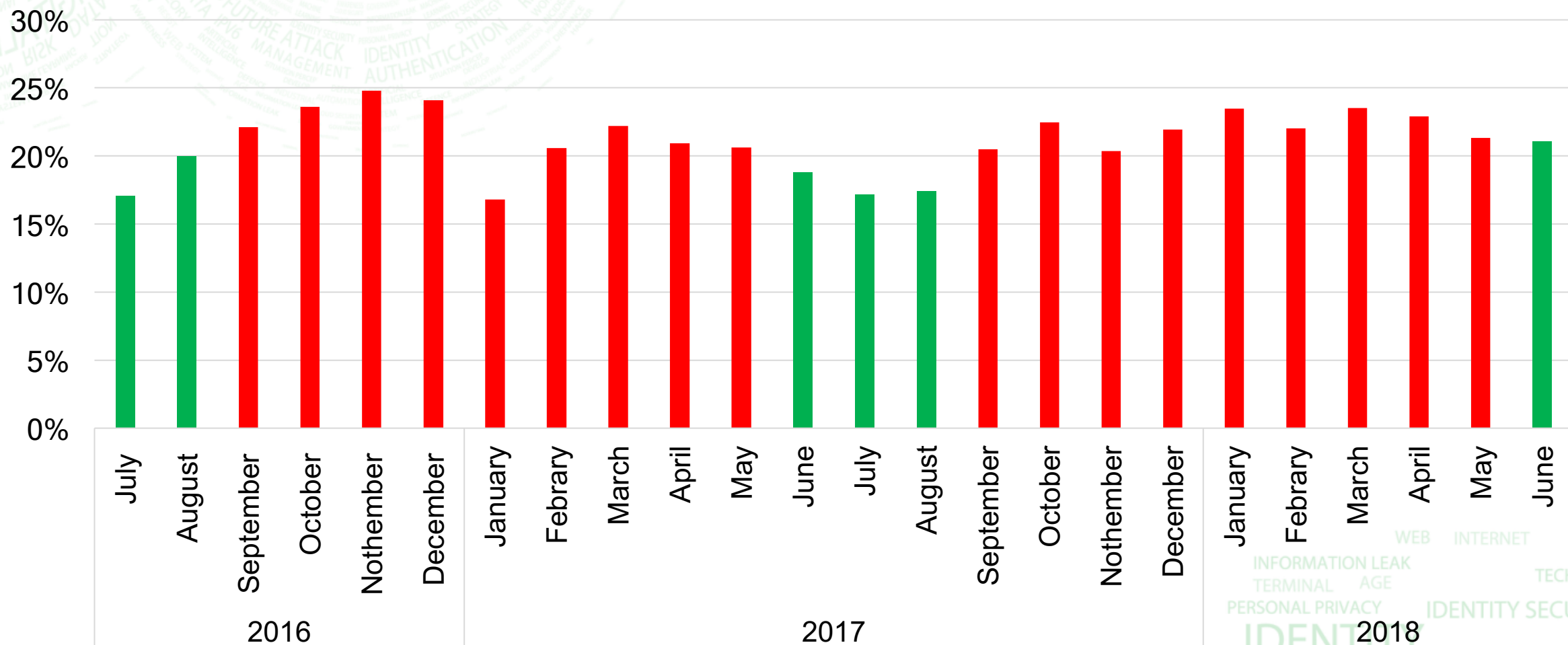
ZERO TRUST SECURITY

KASPERSKY

WEB INTERNET
INFORMATION LEAK
TERMINAL AGE
PERSONAL PRIVACY
IDENTITY SECURITY
IDENTITY
AUTHENTICATION
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China
INDUSTRIAL

KEY ICS RISKS: HUMAN FACTOR

% of ICS computers attacked by malware - according to KSN statistics



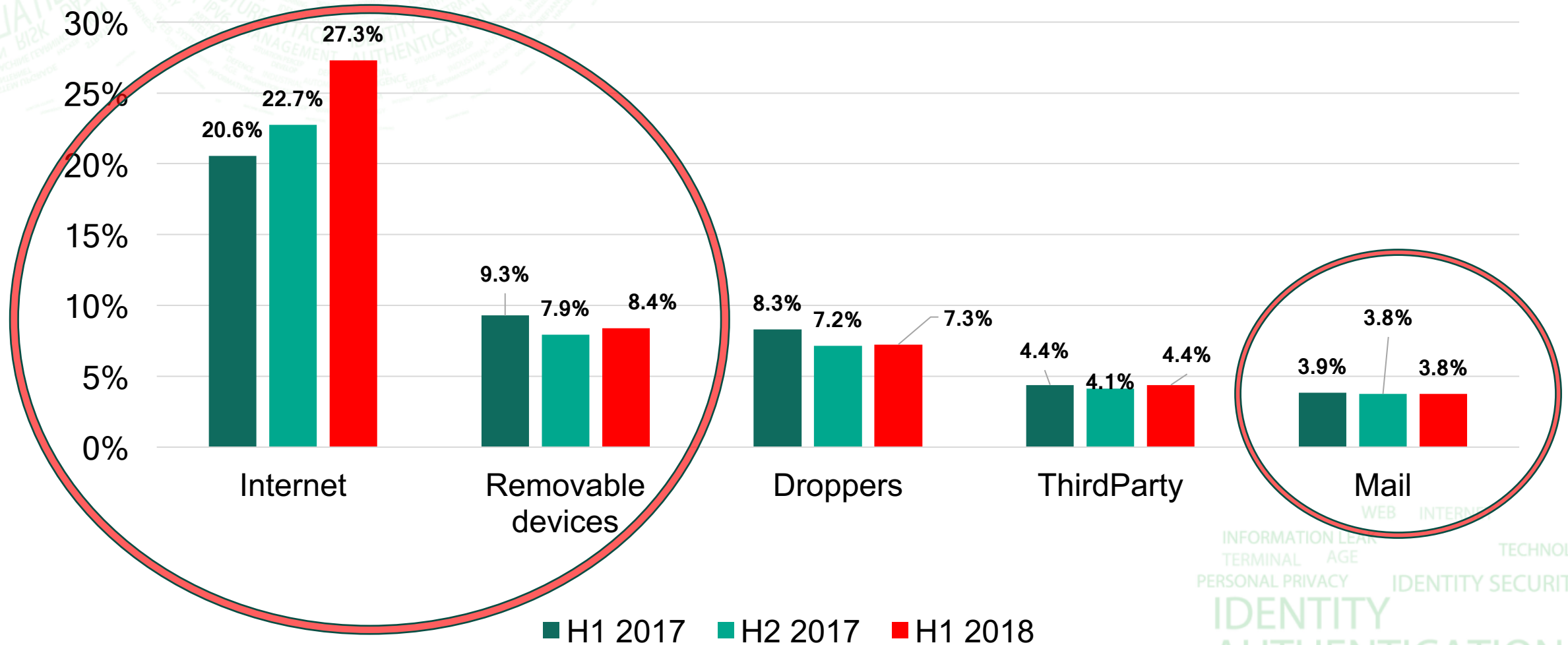
ZERO TRUST SECURITY

KASPERSKY

ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing · China

KEY ICS RISKS: HUMAN FACTOR

% of ICS computers attacked by malware via selected attack vector - according to KSN statistics



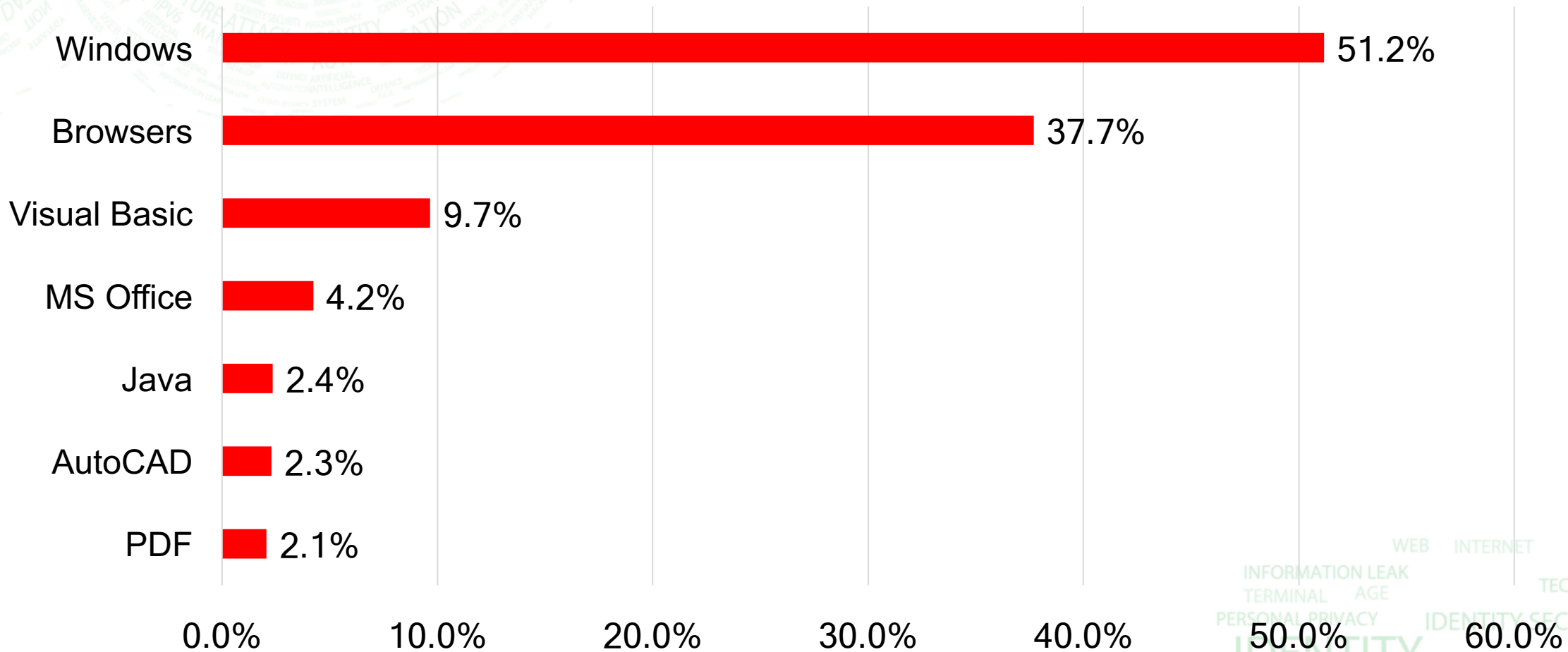
ZERO TRUST SECURITY

KASPERSKY

INFORMATION LEAK
TERMINAL AGE
PERSONAL PRIVACY
IDENTITY SECURITY
IDENTITY
AUTHENTICATION
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China
INDUSTRIAL

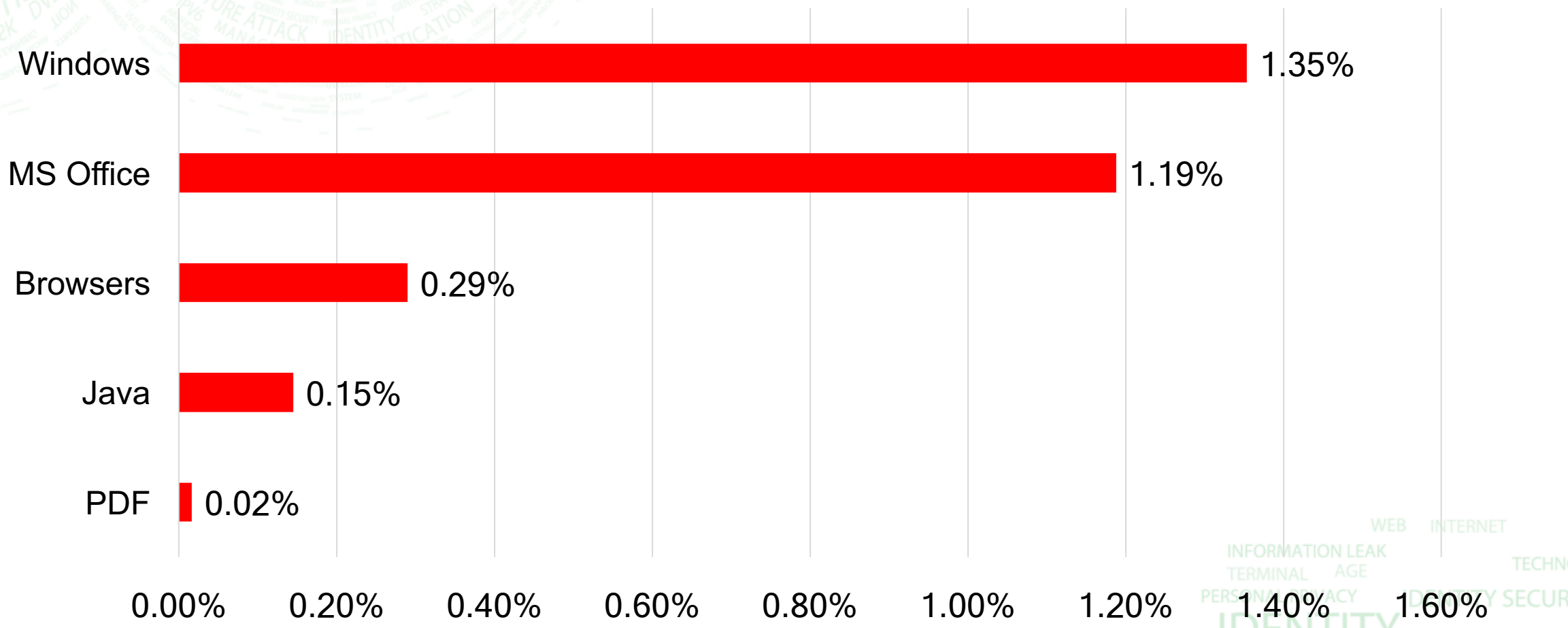
KEY ICS RISKS: NON-ICS SOFTWARE

% of ICS computers attacked by malware run by the selected software platform - according to KSN statistics



KEY ICS RISKS: NON-ICS SOFTWARE

% of ICS computers attacked by exploits for selected software platform - according to KSN statistics



ZERO TRUST SECURITY



WEB INTERNET
INFORMATION LEAK
TERMINAL AGE
PERSONAL PRIVACY
IDENTITY
AUTHENTICATION
TECHNOLOGY
SECURITY
INDUSTRIAL

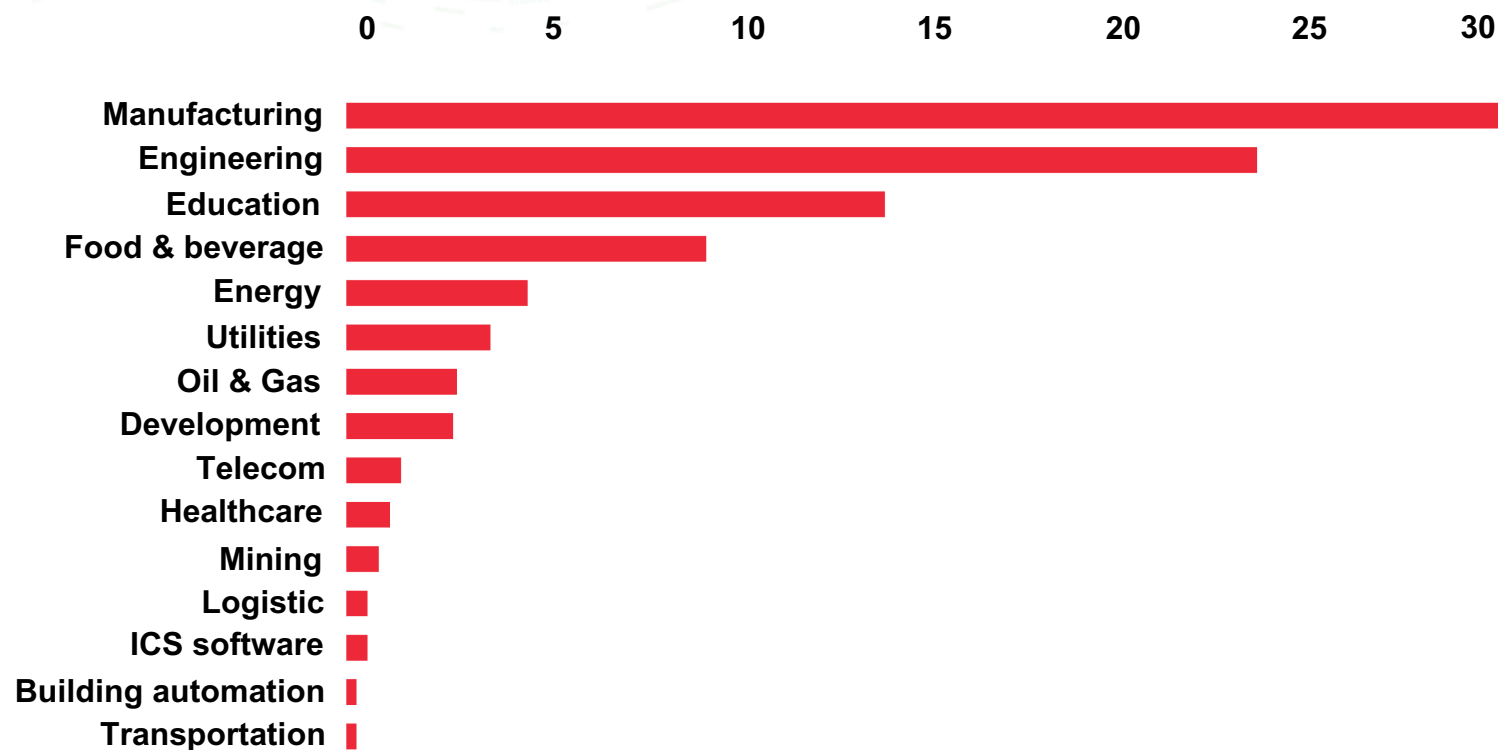
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China

KEY ICS RISKS: INDUSTRIAL SPECIFICS

% of ICS computers attacked by malware - according to KSN statistics



Every 3rd ICS computer under attack was from the manufacturing sector



■ Percentage of all ICs computers attacked

ICS computers in manufacturing companies that produce various materials, equipment and goods accounted for about one third of all attacks

KEY ICS RISKS: GEOGRAPHICAL SPECIFICS – REIGION MATTERS

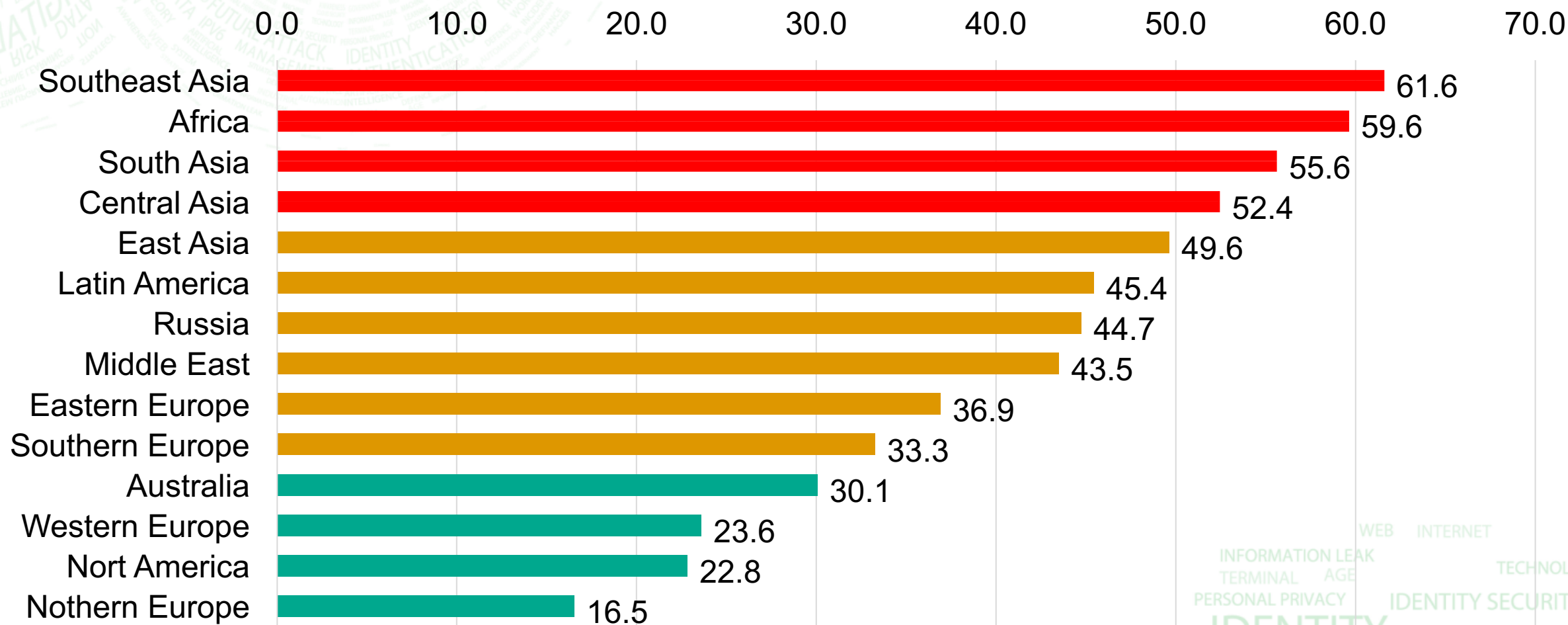
% of ICS computers attacked by malware - according to KSN statistics



ISC 互联网安全大会



360 互联网安全中心



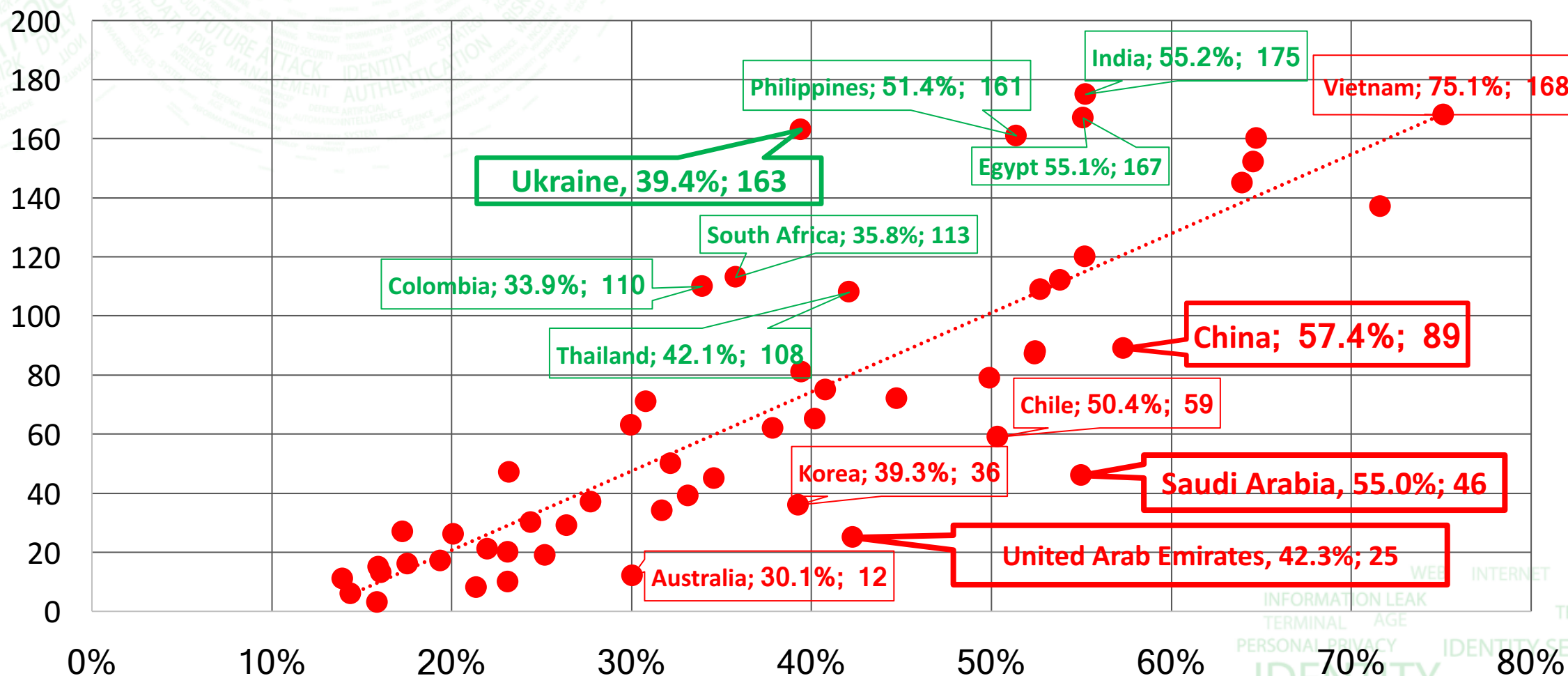
ZERO TRUST SECURITY

KASPERSKY

WEB INTERNET
INFORMATION LEAK
TERMINAL AGE
PERSONAL PRIVACY
IDENTITY SECURITY
IDENTITY
AUTHENTICATION
ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing·China
INDUSTRIAL

KEY ICS RISKS: GDP RANK – MONEY MATTERS

% of ICS computers attacked by malware - according to KSN statistics vs GDP rank





ISC 互联网安全大会



360 互联网安全中心

谢谢！

ISC 互联网安全大会 中国·北京
Internet Security Conference 2018 Beijing · China

ics-cert.kaspersky.com

www.kaspersky.com

KASPERSKY®